GEOS-MPM Verification Suite Report

Generated by verification/runSuite

June 05, 2026

Contents

1 Running and rerunning

Cases are staged under `testRunDirectory/verification/<case>`. The suite records failures and continues unless `--stop-on-error` is supplied. Use `--force` for noninteractive reruns.

```
cd scripts/preProcessing/materialPointMethodParticleFileWriter
verification/runSuite run --python /usr/tce/bin/python3 --force --jobs 4 --case-jobs 2
verification/runSuite report --python /usr/tce/bin/python3
```

2 Timing and dispatch summary

Case	Status	Family	GEOS job(s)	GEOS state	GEOS wall	Issue
pbcCompactionMomentumTest	passed	pbcCompactionMomentumTest	6215927, 6215926	COMPLETED	Elastic: 00:00:50; Elastic-plastic: 00:01:00	

Subcase timing detail. Dispatch time is local wall time spent staging inputs, running PFW, and submitting the GEOS job. GEOS wall and CPU columns are populated from `sacct` when available.

Case	Subcase	GEOS job	State	GEOS wall	Total CPU	NCPUS	Dispatch
pbcCompactionMomentumTest	Elastic	6215927	COMPLETED	00:00:50	00:00:00	6	6.043
pbcCompactionMomentumTest	Elastic-plastic	6215926	COMPLETED	00:01:00	00:00:00	6	6.046

3 Case studies

3.1 pbcCompactionMomentumTest

3.2 PBC compaction momentum conservation

This verification problem compresses a deterministic, periodically repeated array of circular bodies in plane strain. The left and right boundaries are periodic, while the top and bottom boundaries impose equal and opposite vertical compaction. The initial velocity is zero, so the exact total x-momentum is zero for the entire calculation. Any late-time drift in the global particle x-momentum therefore indicates a loss of discrete momentum conservation rather than a physical impulse. The case is intentionally short and severe. It starts with nearly rigid-body motion and then develops large contact and internal forces as the bodies pack and distort. The verification suite uses this transition to separate harmless early-time transport behavior from the late-time momentum drift visible in the deformation movie. The primary metric is

$$P_x(t) = \sum_p m_p v_{p,x}(t), \qquad e_P = \max_t |P_x(t)|, \tag{1}$$

with the expected value $P_x(t) = 0$.

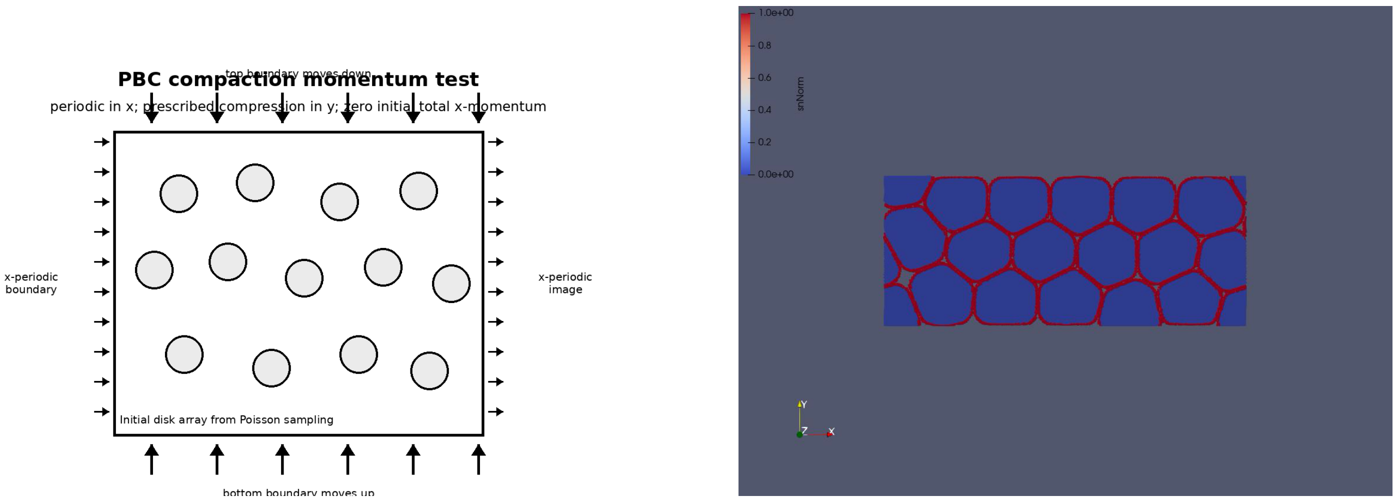


Figure 1: PBC compaction momentum test. Left: schematic of the periodic x-boundaries and symmetric y-compaction. Right: representative late-time elastic deformation used to motivate the x-momentum diagnostic.

Two subcases are run from the same PFW input: an elastic material and an elastic-plastic material. Both use the same geometry, imposed compaction history, and FMPM update. Because the x direction is periodic, the input uses at least three MPI partitions in x; this avoids an under-resolved periodic left/owned/right communication neighborhood while preserving the compact run time of the verification case. The post-processor generates x-velocity contour frames at the initial, quarter, half, three-quarter, and final output states, plus a CSV/plot of total particle x-momentum versus time.

Generated momentum metric. The primary verification metric is the maximum absolute total particle x-momentum. The exact value is zero because the initial x-momentum is zero and the imposed y-compaction does not prescribe net x-impulse. A nonzero drift therefore measures loss of discrete momentum conservation.

Subcase	Samples	Final time	Final P_x	max $\|P_x\|$
Elastic	3587	5.99906	2.62623e-18	2.72914e-18
Elastic-plastic	3391	5.99987	1.84823e-18	1.85839e-18

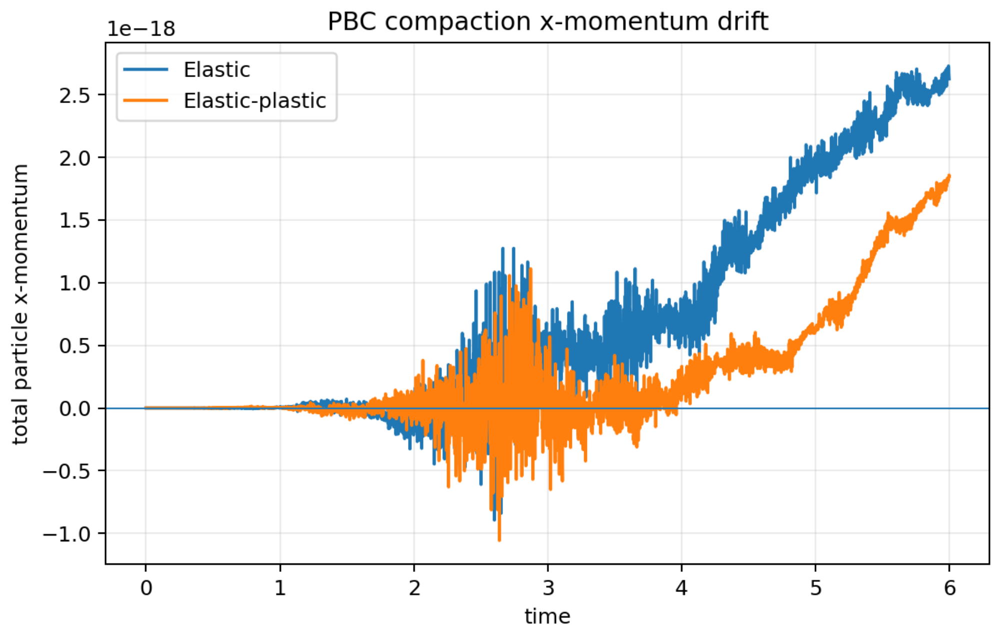


Figure 2: Total particle x-momentum versus time. The exact curve is identically zero.

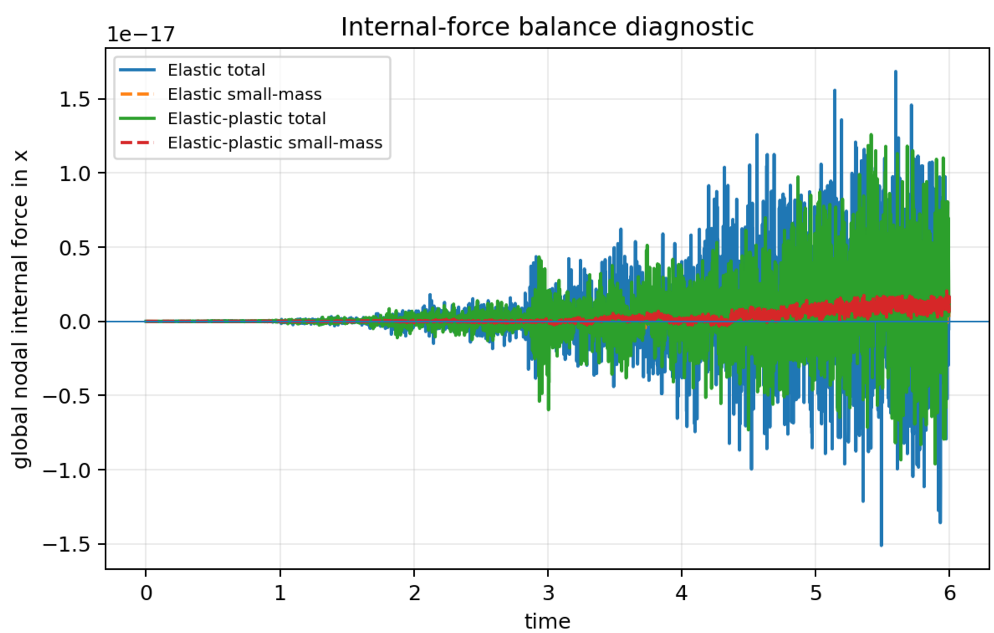
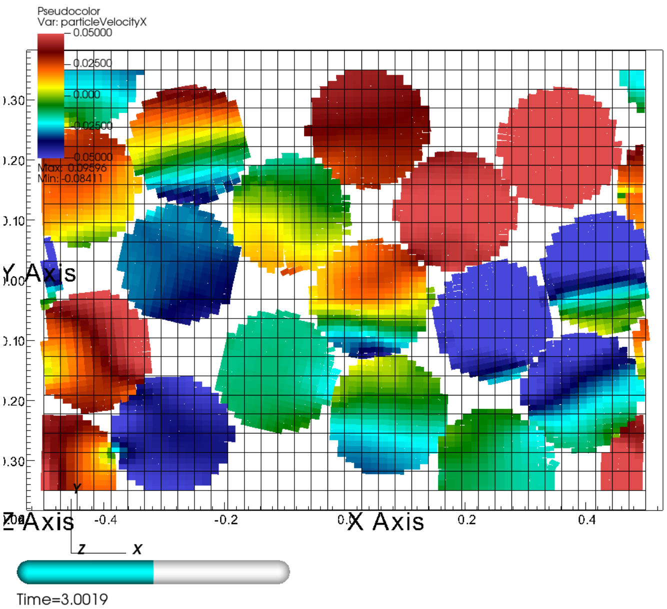
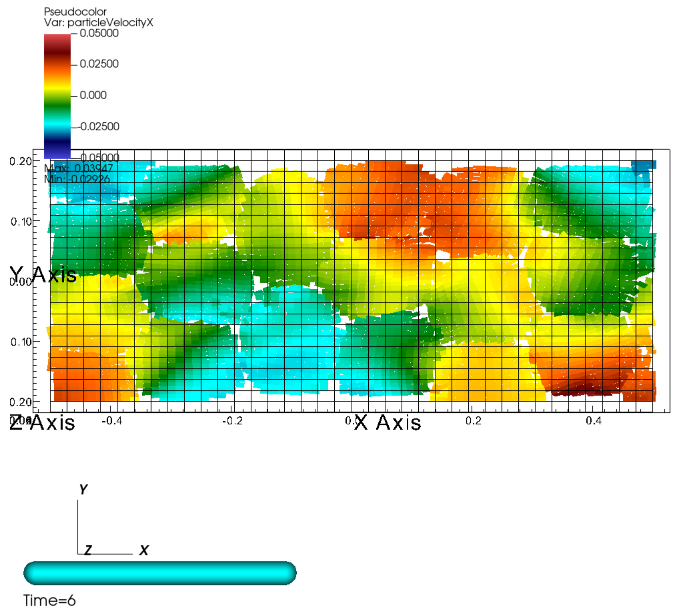


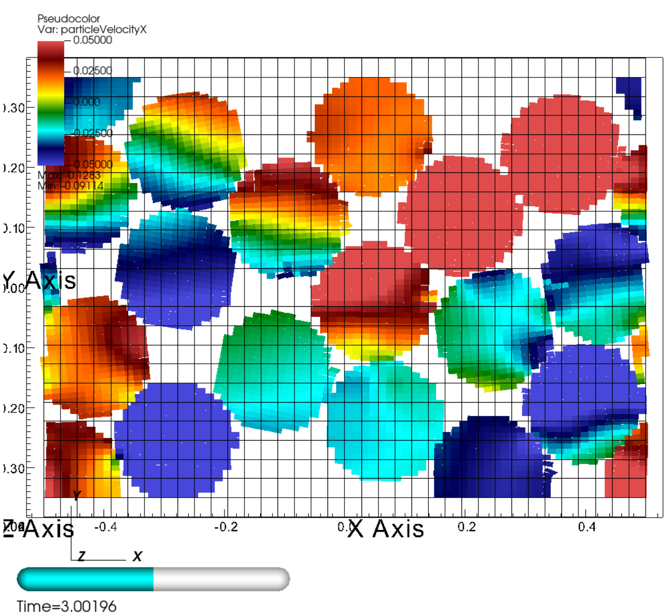
Figure 3: Global x-component of the nodal internal-force balance at the synchronized force-assembly snapshot, including the contribution carried by small-mass nodes.



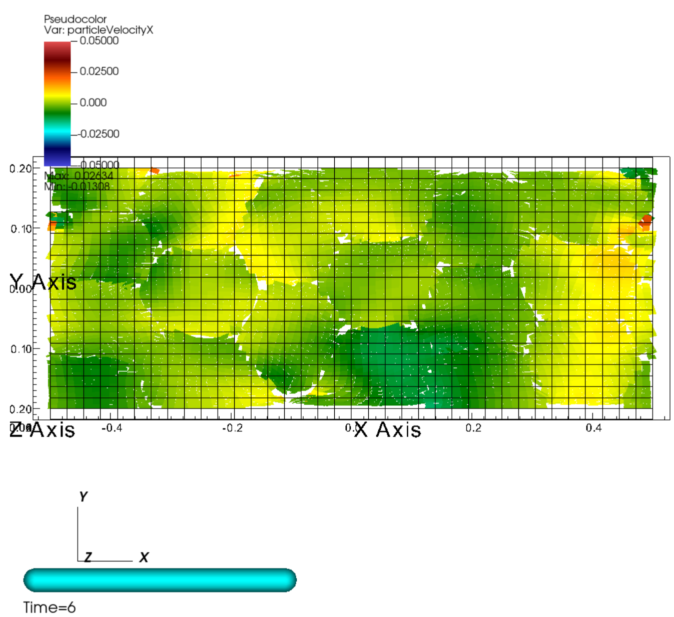
Elastic, $t \approx T/2$



Elastic, $t = T$



Elastic-plastic, $t \approx T/2$



Elastic-plastic, $t = T$

Figure 4: Particle x-velocity contours overlaid on the background grid for the PBC compaction momentum test. The physically expected total x-momentum is zero, so these frames visualize local transverse velocity while the momentum plot quantifies the global conservation error.